

PRACTICE EXAM 1 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 1

Twenty questions. Domains mixed, as they will be on the day.

QUESTIONS 20 PAUSE 9 seconds SCORING By domain

Before you start

- ◆ **Answer every question, including ones you are unsure of.** Guessing is free, and this is practice for that habit.
- ◆ Write your answers down. You cannot score by domain from memory.
- ◆ **Each walkthrough names its domain** so you can total your score per domain afterwards.
- ◆ Do not pause to look things up. A score inflated by notes tells you nothing useful.
- ◆ The scoring guide at the end explains what your pattern of misses means.

ON THE EXAM

Which domain your misses cluster in matters far more than your total. That is the thing you act on.

A company needs to validate uploaded CSV files as soon as they arrive in an Amazon S3 bucket. Validation takes about ninety seconds per file, and files arrive in unpredictable bursts, sometimes none for hours.

The company does not want to manage servers or pay for idle capacity.

Which service should the company use?

- ☐ A Amazon EC2 with an Auto Scaling group
- ☐ B AWS Lambda
- ☐ C AWS Batch
- ☐ D Amazon Lightsail

◆ PAUSE AND DECIDE

Which service should the company use?

- A

 Amazon EC2 with an Auto Scaling group
Servers to manage, and instances billing through the quiet hours.
- B

 AWS Lambda
Event-driven from the upload, ninety seconds is well inside the fifteen-minute limit, and idle costs nothing.
- C

 AWS Batch
For large volumes of queued jobs, not for reacting immediately to a single upload.
- D

 Amazon Lightsail
A simplified server you still manage. It does not respond to events.

THE SIGNAL

Triggered by an upload, unpredictable volume, no servers, and a duration inside fifteen minutes.

A hospital's appointment booking system sees demand rise sharply every winter and fall away in summer. On premises it bought servers for the winter peak, which sat mostly idle the rest of the year.

After migrating, the number of instances now grows and shrinks automatically as demand changes through the year.

Which benefit of the AWS Cloud does this describe?

- | | |
|---|---|
| ☐ A Agility | ☐ B Elasticity |
| ☐ C Economies of scale | ☐ D Operational resilience |

◆ PAUSE AND DECIDE

Which benefit of the AWS Cloud does this describe?

A Agility

Agility is speed of doing something new. Nothing here got faster; the same system runs on a different curve.

C Economies of scale

That is aggregated demand lowering unit prices. No pricing mechanism is described.

B Elasticity

Capacity growing and shrinking automatically as demand changes is the definition.

D Operational resilience

Needs availability, fault tolerance or security. Nothing has failed.

THE SIGNAL

Capacity tracking demand is elasticity. Time to launch would be agility.

A company runs an image-processing function on AWS Lambda. A security audit asks who patches the operating system underneath the function, and who is responsible for vulnerabilities in the function's own code.

The company is unsure how the shared responsibility model applies to serverless.

Which statement is correct?

- ☐ A AWS patches the OS; the customer is responsible for the function code
- ☐ B The customer patches the OS; AWS is responsible for the function code
- ☐ C AWS is responsible for both
- ☐ D The customer is responsible for both

◆ PAUSE AND DECIDE

Which statement is correct?

- A

 AWS patches the OS; the customer is responsible for the function code
With Lambda there is no customer-accessible OS, so AWS patches it. Your code and permissions remain yours.
- B

 The customer patches the OS; AWS is responsible for the function code
Reversed. You cannot reach a Lambda operating system, and AWS never owns your code.
- C

 AWS is responsible for both
Serverless does not make you responsible for nothing. Your code is always yours.
- D

 The customer is responsible for both
That describes EC2, where you do patch the guest OS. Lambda is not EC2.

THE SIGNAL

Managed services take the operating system away from you. They never take your code, data or permissions away.

A company runs a steady production workload on EC2 and wants a substantial discount over three years. During that period it plans to migrate some components to AWS Fargate and rewrite others as AWS Lambda functions.

It wants the discount to keep applying as those migrations happen.

Which purchase option should the company choose?

- | | |
|--|--|
| ☐ A Standard Reserved Instances | ☐ B Compute Savings Plans |
| ☐ C EC2 Instance Savings Plans | ☐ D Spot Instances |

◆ PAUSE AND DECIDE

Which purchase option should the company choose?

- A

 Standard Reserved Instances
Locked to instance attributes and to EC2. The commitment would strand as workloads move.
- B

 Compute Savings Plans
Applies automatically across EC2, Fargate and Lambda, so the commitment follows the architecture.
- C

 EC2 Instance Savings Plans
Deeper discount, but committed to an instance family in one Region and limited to EC2.
- D

 Spot Instances
No commitment discount, and interruption is unsuitable for steady production.

THE SIGNAL

Fargate and Lambda named alongside EC2 leaves exactly one option that covers all three.

A company stores rendered video thumbnails in Amazon S3. They are accessed infrequently but must be available in milliseconds when requested. If they were ever lost, the company could regenerate every one from the source video within an hour. It wants the lowest storage cost consistent with those facts.

Which storage class should the company use?

- ☐ A S3 Standard
- ☐ B S3 Standard-IA
- ☐ C S3 One Zone-IA
- ☐ D S3 Glacier Deep Archive

◆ PAUSE AND DECIDE

Which storage class should the company use?

- A

 S3 Standard
For frequent access. The data is explicitly infrequently accessed, so this overpays.
- B

 S3 Standard-IA
The near miss. Correct access pattern, but it stores across three AZs, which recreatable data does not need.
- C

 S3 One Zone-IA
Infrequent access with millisecond retrieval, in a single AZ. Recreatable data is exactly its use case.
- D

 S3 Glacier Deep Archive
Retrieval takes up to forty-eight hours. Milliseconds is required.

THE SIGNAL

Could regenerate them is the clause that licenses One Zone. Without it, Standard-IA would be correct.

A company's finance director needs to change the AWS account's registered email address and downgrade its AWS Support plan. An administrator with the AdministratorAccess policy attached attempts both and finds neither is possible.

The administrator asks why.

Why can the administrator not perform these tasks?

- ☐ A AdministratorAccess excludes billing permissions by default
- ☐ B These are root user tasks that no IAM policy can grant
- ☐ C A service control policy is blocking the actions
- ☐ D The administrator needs multi-factor authentication enabled

◆ PAUSE AND DECIDE

Why can the administrator not perform these tasks?

- A

 AdministratorAccess excludes billing permissions by default
Plausible-sounding and untrue. AdministratorAccess grants broad permissions; the limit here is elsewhere.
- B

 These are root user tasks that no IAM policy can grant
Changing the account email and changing the support plan are both reserved to the root user.
- C

 A service control policy is blocking the actions
Possible in principle, but nothing in the scenario mentions an organisation or an SCP.
- D

 The administrator needs multi-factor authentication enabled
MFA affects sign-in assurance, not whether an action is reserved to the root user.

THE SIGNAL

Both tasks change the account itself rather than a resource in it. Account-level actions need root.

A company is redesigning a workload to reduce the energy it consumes. It plans to move to more efficient instance types, shut down non-production environments overnight, and choose Regions with lower carbon intensity.

It wants to know which pillar of the Well-Architected Framework this work belongs to.

Which pillar does this represent?

- | | |
|--|---|
| ☐ A Cost Optimization | ☐ B Performance Efficiency |
| ☐ C Sustainability | ☐ D Operational Excellence |

◆ PAUSE AND DECIDE

Which pillar does this represent?

- A

Cost Optimization
The strongest distractor, since every listed action also saves money. But the stated goal is energy, not spend.
- B

Performance Efficiency
About using the right resources as demand changes. Efficiency here is measured in energy, not performance.
- C

Sustainability
Minimising the environmental impact of running a workload. The stated goal is energy consumption.
- D

Operational Excellence
Process, runbooks and observability. Nothing about how the team operates is changing.

THE SIGNAL

Every action here reduces cost too. The goal decides the pillar, and the stated goal is energy.

An application on EC2 instances in a private subnet writes large volumes of data to Amazon S3. Compliance requires the traffic never to traverse the public internet, and the finance team has queried the NAT gateway processing charges.

The company wants to satisfy both concerns.

What should the company implement?

- ☐ A A VPC gateway endpoint for Amazon S3
- ☐ B An additional NAT gateway
- ☐ C AWS Direct Connect
- ☐ D An internet gateway attached to the private subnet

◆ PAUSE AND DECIDE

What should the company implement?

- A

 A VPC gateway endpoint for Amazon S3
Traffic stays on the AWS network, satisfying compliance, and gateway endpoints carry no charge.
- C

 AWS Direct Connect
Connects an on-premises network to AWS. Both ends here are already inside AWS.
- B

 An additional NAT gateway
Increases cost and still routes traffic out through the internet path.
- D

 An internet gateway attached to the private subnet
Makes the subnet public and puts the traffic on the internet, breaking the compliance requirement.

THE SIGNAL

Both stated concerns point the same way. Endpoints keep traffic private, and gateway endpoints are free.

A company pushes container images to Amazon ECR. Security wants to know continuously whether any image contains software with known published vulnerabilities, and to be told when a newly published vulnerability affects an image already stored.

Nothing should need installing into the images themselves.

Which service meets this requirement?

- | | |
|---|---|
| ☐ A Amazon GuardDuty | ☐ B Amazon Inspector |
| ☐ C Amazon Macie | ☐ D AWS Config |

◆ PAUSE AND DECIDE

Which service meets this requirement?

- A

 Amazon GuardDuty
Detects active threats from behaviour. A stored image is not exhibiting behaviour.
- B

 Amazon Inspector
Scans ECR images for known vulnerabilities and rescans as new ones are published.
- C

 Amazon Macie
Finds sensitive data inside Amazon S3. It does not scan images for vulnerabilities.
- D

 AWS Config
Evaluates resource configuration against rules. It has no vulnerability database.

THE SIGNAL

Known published vulnerabilities is the Inspector signature. Nothing else in AWS scans for them.

A company must migrate a Microsoft SQL Server database to a managed AWS database service. The application cannot be modified and requires SQL Server specifically.

The company also wants automated backups and the option of automatic failover to another Availability Zone.

Which service should the company use?

- | | |
|--|--|
| ☐ A Amazon Aurora | ☐ B Amazon RDS |
| ☐ C Amazon DynamoDB | ☐ D Amazon Redshift |

◆ PAUSE AND DECIDE

Which service should the company use?

- A

 Amazon Aurora
The key trap. Aurora is MySQL and PostgreSQL compatible only. It cannot run SQL Server.
- B

 Amazon RDS
Supports SQL Server as a managed engine, with automated backups and optional Multi-AZ failover.
- C

 Amazon DynamoDB
Key-value NoSQL. The application expects a relational SQL Server database.
- D

 Amazon Redshift
An analytical warehouse, not a transactional database for an application.

THE SIGNAL

Aurora is only ever MySQL or PostgreSQL compatible. Naming SQL Server or Oracle means standard RDS.

A company runs several hundred virtual machines on VMware in its own data centre. It wants to move the whole estate to AWS quickly, without re-buying hardware, and without changing how its operations team works day to day.

No applications will be modified or re-architected.

Which migration strategy does this describe?

- | | |
|-------------------------------------|---------------------------------------|
| ☐ A Rehost | ☐ B Replatform |
| ☐ C Relocate | ☐ D Refactor |

◆ PAUSE AND DECIDE

Which migration strategy does this describe?

A Rehost

Close, but rehost moves individual servers onto EC2. This moves an entire virtualisation estate wholesale.

C Relocate

Moving infrastructure wholesale without changing applications or operations. Exactly this scenario.

B Replatform

Requires a component to be swapped for a managed service. Nothing is being swapped.

D Refactor

Rewriting applications, which the final sentence explicitly rules out.

THE SIGNAL

Whole estate, no hardware re-purchase, operations unchanged. That combination is Relocate, not Rehost.

A company's public website suffered an outage when its TLS certificate expired unnoticed. The certificate was bought from a third party and installed manually on an Application Load Balancer.

The company wants renewal to happen without anyone remembering to act.

What should the company use?

- ☐ A AWS Key Management Service
- ☐ B AWS Secrets Manager
- ☐ C AWS Certificate Manager
- ☐ D AWS Systems Manager Parameter Store

◆ PAUSE AND DECIDE

What should the company use?

- A

AWS Key Management Service
Manages encryption keys for data at rest. Certificates and TLS are a different problem.
- B

AWS Secrets Manager
Stores and rotates credentials. Storing a certificate more securely does not stop it expiring.
- C

AWS Certificate Manager
Issues and automatically renews TLS certificates, and deploys them to the load balancer.
- D

AWS Systems Manager Parameter Store
Configuration storage. It has no certificate lifecycle at all.

THE SIGNAL

Without anyone remembering to act rules out every option that only stores the certificate better.

An operations team receives CloudWatch alarms for CPU and network on its EC2 fleet, but has no visibility of memory usage or free disk space inside the instances. It wants those metrics in the same dashboards.

The instances run a standard Amazon Linux image.

What must the team do?

- ☐ A Enable detailed monitoring on the instances
- ☐ B Install the CloudWatch agent on the instances
- ☐ C Enable AWS CloudTrail data events
- ☐ D Attach an IAM role granting CloudWatch read access

◆ PAUSE AND DECIDE

What must the team do?

- A

 Enable detailed monitoring
Increases how often existing metrics are reported. It does not add metrics that were never collected.
- B

 Install the CloudWatch agent
Memory and disk usage live inside the operating system, which the hypervisor cannot see. The agent reports them.
- C

 Enable AWS CloudTrail data events
Records API activity. It has no view of instance resource usage.
- D

 Attach an IAM role granting CloudWatch read access
Permissions to read metrics do not cause metrics to be produced.

THE SIGNAL

Memory and disk space are inside the OS. AWS stops at the hypervisor, so something must report from within.

A company runs a business-critical trading platform on AWS. It needs a fifteen-minute response if that platform goes down, and a named AWS contact who understands its architecture and can advise proactively.

Budget is available.

Which support plan meets these requirements?

- | | |
|---|---|
| ☐ A AWS Developer Support | ☐ B AWS Business Support |
| ☐ C AWS Enterprise On-Ramp | ☐ D AWS Enterprise Support |

◆ PAUSE AND DECIDE

Which support plan meets these requirements?

- A

 AWS Developer Support
Business hours only, and it does not cover production severities at all.
- B

 AWS Business Support
24/7 with one hour for production down, but no named contact and no fifteen-minute severity.
- C

 AWS Enterprise On-Ramp
The near miss. Thirty minutes rather than fifteen, and a pool of TAMs rather than a named one.
- D

 AWS Enterprise Support
Fifteen-minute business-critical response and a designated Technical Account Manager.

THE SIGNAL

Fifteen minutes and a named contact both point at Enterprise. Enterprise On-Ramp offers weaker versions of each.

A company beginning its cloud programme has discovered that it has no standard way to build environments, no landing zone, and no agreed approach to modernising its applications. Its engineers each set things up differently.

It wants to know which AWS Cloud Adoption Framework perspective addresses this.

Which perspective is it?

- | | |
|---------------------------------------|-------------------------------------|
| ☐ A Platform | ☐ B People |
| ☐ C Governance | ☐ D Business |

◆ PAUSE AND DECIDE

Which perspective is it?

A Platform

Building the cloud environment and modernising workloads. A landing zone and build standards are Platform capabilities.

C Governance

Cost, risk and compliance oversight. No governance concern is described.

B People

Skills, culture and change. The problem is missing technical standards, not missing skills.

D Business

Aligning investment with business objectives. Nothing about business outcomes appears here.

THE SIGNAL

Landing zone and modernising applications are Platform capabilities. Ask what the missing thing actually is.

An insurance claim is processed in seven steps. Some steps run in parallel, one branches depending on the claim value, and any step that fails must be retried three times before the claim is routed for manual review.

The company wants to see where a claim stopped when something goes wrong.

Which service should coordinate this?

- ☐ A Amazon SNS
- ☐ B Amazon SQS
- ☐ C AWS Step Functions
- ☐ D Amazon EventBridge

◆ PAUSE AND DECIDE

Which service should coordinate this?

- A

 Amazon SNS
Broadcasts a message to subscribers. It has no concept of steps, order or retries.
- B

 Amazon SQS
Buffers work for consumers. It decouples components but does not orchestrate a sequence.
- C

 AWS Step Functions
Steps, parallel branches, conditional routing, retries and a visual view of where a run stopped.
- D

 Amazon EventBridge
Routes events to targets by rule. Routing is not the same as coordinating a multi-step workflow.

THE SIGNAL

Order, branching, retries and visibility of where it stopped. Four orchestration signals in one scenario.

A company is expanding from one AWS account to about forty. It wants a secure multi-account environment established quickly, with a standard set of controls applied from the outset, centralised logging, and a repeatable way for teams to request new accounts. It has no existing multi-account design and little time to create one.

Which service should the company use?

- | | |
|---|--|
| ☐ A AWS Organizations with hand-written SCPs | ☐ B AWS Control Tower |
| ☐ C AWS Config with conformance packs | ☐ D AWS IAM Identity Center |

◆ PAUSE AND DECIDE

Which service should the company use?

- A

 AWS Organizations with hand-written SCPs
Would work eventually, but the company must design everything itself. The stem rules that out on time.
- B

 AWS Control Tower
A landing zone with controls, centralised logging and Account Factory for provisioning new accounts.
- C

 AWS Config with conformance packs
Detects non-compliance. It builds no accounts and establishes no environment.
- D

 AWS IAM Identity Center
Single sign-on across accounts. It does not create accounts or apply controls.

THE SIGNAL

Quickly, from the outset, and a repeatable way to request accounts. Account Factory is the deciding capability.

A company runs its entire platform in a single AWS Region, deployed across three Availability Zones. Its regulator now requires the company to demonstrate it can continue operating if that entire Region became unavailable for an extended period.

Availability within the Region has never been a problem.

What should the company do?

- ☐ A Add a fourth Availability Zone
- ☐ B Deploy a disaster recovery environment in a second Region
- ☐ C Use Amazon CloudFront in front of the application
- ☐ D Increase the size of the instances

◆ PAUSE AND DECIDE

What should the company do?

- A

 Add a fourth Availability Zone
More resilience inside the Region that is itself the stated risk. It does not address Region loss.
- B

 Deploy a disaster recovery environment in a second Region
Only a second Region survives the loss of the first. That is what the regulator is asking for.
- C

 Use Amazon CloudFront
Caches content near users. If the origin Region is gone, uncached requests still fail.
- D

 Increase the size of the instances
Capacity, not resilience. Larger instances in a lost Region are still lost.

THE SIGNAL

Multi-AZ survives a data centre. Only multi-Region survives a Region. The requirement names the Region.

A finance team is comparing the cost of its on-premises estate with running the same workloads on AWS. It lists the data centre lease, the three-yearly server refresh, its security operations team, and the electricity bill for cooling.

It wants to know which of these disappear entirely after migrating.

Which two costs are eliminated? (Choose two.)

- ☐ A The data centre lease
- ☐ B The server refresh
- ☐ C The security operations team
- ☐ D Staff training
- ☐ E Application licensing

◆ PAUSE AND DECIDE

Which two costs are eliminated? (Choose two.)

- A

 The data centre lease
AWS owns the facilities. The lease, power and cooling genuinely go away.
- B

 The server refresh
Hardware purchase becomes consumption-based charging. The refresh cycle disappears.
- C

 The security operations team
Securing your data stays yours under the shared responsibility model. It moves, it does not vanish.
- D

 Staff training
Training usually increases during a migration, as people learn AWS.
- E

 Application licensing
Licences still apply, whether brought with you or included in the hourly rate.

THE SIGNAL

Eliminated means gone entirely. Only facilities and hardware purchase qualify; everything else changes shape.

A contractor needs temporary access to read objects in one Amazon S3 bucket, for six weeks. A junior administrator proposes attaching the AdministratorAccess policy because it is quicker than writing something specific.

The security team objects.

What should be done instead?

- ☐ A Attach AdministratorAccess and remove it after six weeks
- ☐ B Grant only read access to that one bucket, following least privilege
- ☐ C Share the root user credentials for the duration
- ☐ D Add the contractor to the administrators group

◆ PAUSE AND DECIDE

What should be done instead?

- A

 Attach AdministratorAccess and remove it after six weeks
Grants full control of the account for six weeks to read one bucket. Convenience is never the justification.
- B

 Grant only read access to that one bucket, following least privilege
The minimum permission required for the task, and nothing more.
- C

 Share the root user credentials for the duration
Root credentials are never shared, and no IAM policy can restrict the root user.
- D

 Add the contractor to the administrators group
The same excessive grant as option A, arranged through a group.

THE SIGNAL

When two options both work and one grants less access, the one granting less is correct. Every time.

Scoring guide

DOMAIN	QUESTIONS	READ THIS IF YOU MISSED TWO OR MORE
D1 Cloud Concepts	2, 7, 11, 15, 19	Rewatch modules 1 to 4. These are vocabulary, not reasoning.
D2 Security	3, 6, 9, 12, 17, 20	Rewatch modules 5 to 9. Start with the shared responsibility model.
D3 Technology	1, 5, 8, 10, 13, 16, 18	Rewatch the specific module, not all eight. Check which service family.
D4 Billing	4, 14	Rewatch modules 18 to 20. Two questions is a small sample; do not over-read it.

ON THE EXAM

Sixteen or more out of twenty is a comfortable position. Below twelve, work the weak domain before taking exam two.

What to do next

- ◆ **Score by domain, not by total.** Two misses in one domain matters more than five spread evenly.
- ◆ Rewatch only your weakest domain. Revising what you already know moves nothing.
- ◆ Do not take exam two immediately. The gap is what makes the second score meaningful.
- ◆ Review any question you guessed correctly. A lucky guess is a gap you cannot see in the score.
- ◆ **Reread the walkthroughs for questions you got right but found hard.** Those are the fragile ones.

ON THE EXAM

Exam two covers different topics entirely. It is not a retest of this paper.